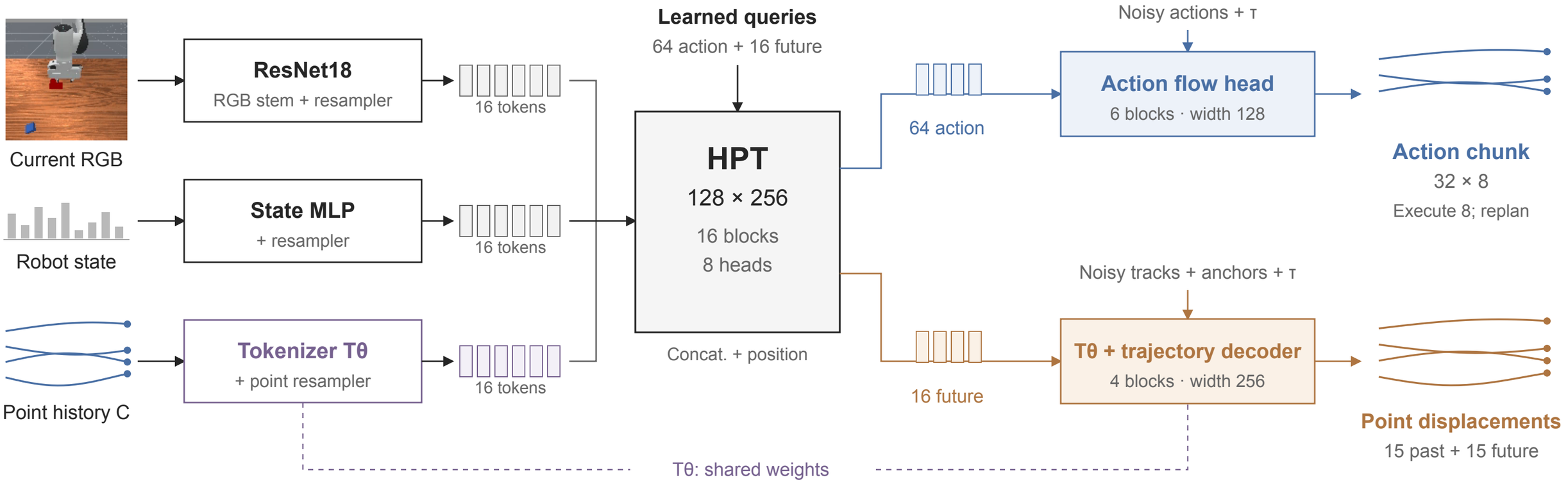
4D MAE / B5

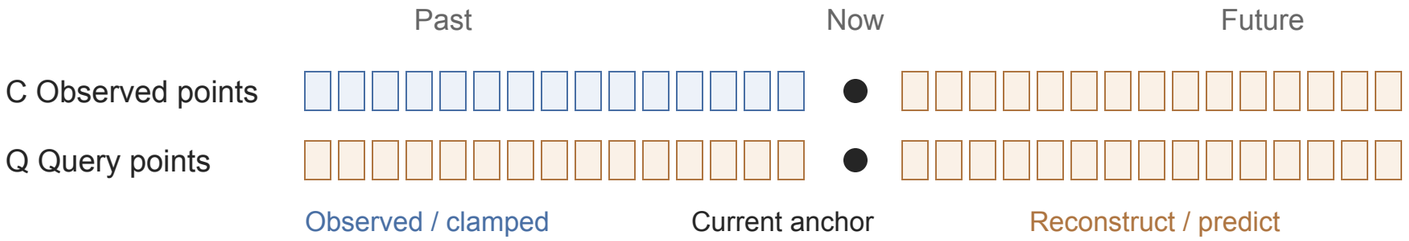
PullCube configuration

(a) Policy architecture



Control uses the action branch. Trajectory completion supplies auxiliary training supervision.

(b) Trajectory supervision



$$L = L_{action} + (L_{C,future} + L_{Q,past} + L_{Q,future}) / 3$$

Flow-matching losses on valid targets.

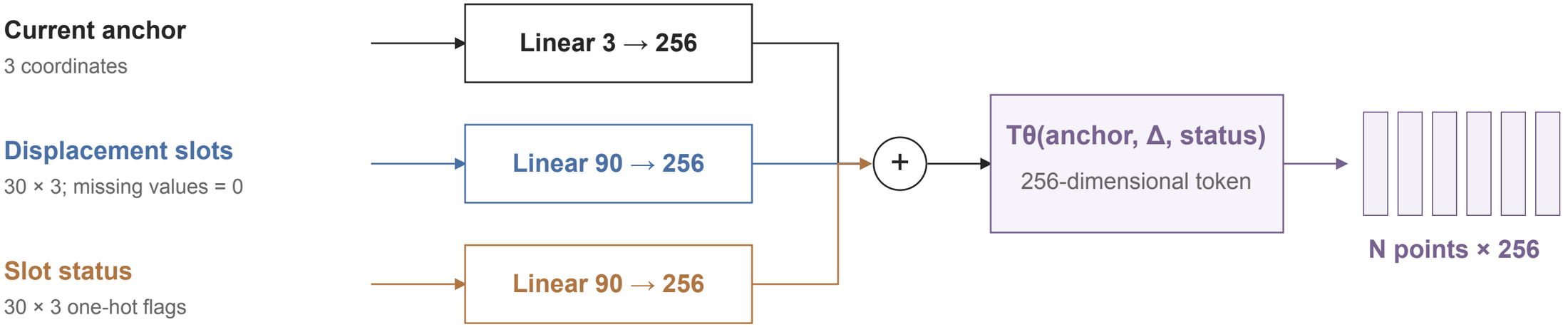
One decoder, called separately for C and Q.

64 C points · 64 Q points · 16 history frames · privileged simulator history in PullCube

All trajectory sketches are schematic.

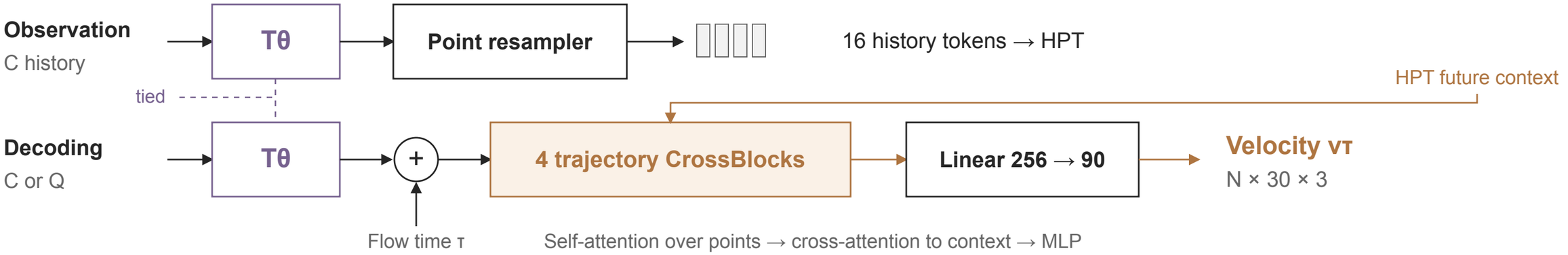
B5: shared trajectory tokenizer

(a) One trajectory → one point token



Status: missing / observed / noisy. All 30 time slots are packed into each point token.

(b) The same Tθ in both paths



Sampling: 50 Euler steps. Observed C past is clamped at every step.

τ is generative time; the 30 output slots are physical time. Add the current anchor to recover absolute XYZ.